

Project Title: E-Commerce Web Application

Version: 1.0

Prepared By: Venkatesh

Date: 20 Apr 2025

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for a web-based e-commerce platform. The platform will allow users to browse products, add items to a shopping cart, place orders, and manage their profiles. Admin users will manage inventory, process orders, and view analytics.

1.2 Scope

This platform will support:

- User registration/login
- Product browsing and search
- Shopping cart management
- Order placement and history
- Payment gateway integration (simulated/demo)
- Admin panel for inventory and order management

1.3 Intended Audience

- Project team members
- Academic supervisors
- Future developers/maintainers

1.4 Definitions

- **User:** A registered customer
- **Admin:** System administrator with management privileges
- **Product:** An item listed for sale
- **Cart:** Temporary collection of products selected by a user

2. Overall Description

2.1 Product Perspective

The system will be a standalone web application. It will follow a client-server architecture and support responsive design.

2.2 Product Functions

- User authentication
- Product catalog with filtering and search
- Shopping cart and order checkout
- Basic payment processing (simulated)
- Admin dashboard for CRUD operations on products and viewing orders

2.3 User Classes and Characteristics

- **Customer:** Can browse, search, purchase products, and view order history.
- **Admin:** Can add/edit/delete products, manage users, and view all orders.

2.4 Operating Environment

- Web browser (Chrome, Firefox, Edge)
- Backend: Node.js / Django / PHP
- Frontend: HTML/CSS/JS or React
- Database: MySQL / MongoDB

2.5 Assumptions and Dependencies

- Payment gateway is simulated, not live.
- Email notifications may be simulated using console output or logs.

3. Specific Requirements

3.1 Functional Requirements

- **FR1:** The system shall allow users to register and log in.
- **FR2:** Users shall be able to browse and search products.

- **FR3:** Users shall be able to add items to the shopping cart.
- **FR4:** Users shall be able to place an order.
- **FR5:** Admins shall be able to add, update, and delete products.
- **FR6:** Admins shall be able to view all user orders.

3.2 Non-Functional Requirements

- **NFR1:** The application shall respond to user actions within 2 seconds.
- **NFR2:** The system shall be available 99% of the time.
- **NFR3:** Passwords shall be stored using secure hashing (e.g., bcrypt).
- **NFR4:** The user interface shall be mobile responsive.

3.3 Interface Requirements

- Web-based GUI
 - REST API for backend communication
 - Admin panel accessible after login with admin credentials
-

4. System Models (Optional for Students)

You can include:

- Use case diagrams
 - ER diagrams for database
 - Basic flowcharts
-

5. Appendix

- Tools: VS Code, MySQL Workbench, GitHub
- Technologies: React, Springboot